

Final Term Project 1

Pac-Dual: A Split-Screen Networked Maze Game Using Socket Programming

This project involves the development of a real-time, two-player Pac-Man game using Python. The game is set in a vertically split screen where each half represents a player's home territory, but the maze is physically continuous. This means players can navigate freely from their own half into the opponent's half, creating a dynamic and connected environment.

Each player controls a single Pac-Man character. While in their own half, the player acts as a traditional Pac-Man—collecting dots and power pellets while avoiding ghosts. However, when a player crosses into the other side of the maze, their Pac-Man automatically transforms into a ghost. In this ghost form, the player is no longer vulnerable but instead gains the ability to chase and attempt to capture the opponent's Pac-Man.

Each half of the maze contains independently moving ghosts that randomly roam and serve as additional hazards. If a player, while in Pac-Man mode, is caught by any ghost they lose. Consuming power pellets temporarily freezes all ghosts in the player's current half, including the opponent if they are acting as a ghost, creating a strategic opportunity for escape or aggressive play.

The game concludes when a player successfully collects all the dots in their half of the maze or captures the opponent while acting as a ghost. This dynamic switching between Pac-Man and ghost modes based on location encourages both offensive and defensive strategies, enhancing gameplay depth and competitiveness.

The game must be implemented as a real-time networked application using socket programming to enable communication between two human players over separate machines. The design must follow object-oriented programming principles to ensure modularity, scalability, and maintainability of the game components.